BYTEHAMR.COM

# Disk Hamr

## OWNER'S GUIDE

*Replacement controller board for the Apple UniDisk 3.5, model A2M2053.*

REV 1.0, FIRMWARE V1



# Contents

*Disk Hamr rev 1.0, firmware v1*


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SECTION ONE

# 1 What this is

## Which drive this is for: the UniDisk 3.5, model A2M2053

Apple sold several 3.5-inch drives that look alike and share the same Sony mechanism. Disk Hamr fits exactly one of them. Check before you buy or open anything:

- **Front badge** reads "UniDisk 3.5". Beige case, matching the Apple IIe.
- **Label on the underside** reads model **A2M2053**.
- **Rear panel** has a DB-19 socket for the next drive in the chain, next to the captive cable.
- **Inside**, the controller board's silk reads **IF-83**, and it carries a 65C02, ROM, RAM and a large Sony gate array marked CXD1013.

It is **not** for:

- the Apple 3.5 Drive, model A9M0106 (platinum, badge "Apple 3.5 Drive", for the IIgs, IIc Plus and Macintosh). Same mechanism, different board, different protocol.
- the Macintosh 800K External Drive, model M0131, or any Macintosh-only drive.
- the Disk II, UniDisk 5.25 or any 5.25-inch drive.




**FIG 1-1** *The underside label with A2M2053 circled, and the front badge.*

# 1 What this is, continued

The UniDisk 3.5 is Apple's 800K floppy drive for the Apple II. Inside it are a Sony mechanism and a controller board, Apple part IF-83 (service part 661-0307). When the controller board fails there is no repair path: it is built around a Sony gate array with no datasheet and no replacement.

Disk Hamr replaces the whole controller board. The mechanism, the case, the front panel, the LED, the eject button and the drive's own cables stay. The board speaks SmartPort to the Apple and drives the Sony mechanism directly. The firmware is new code on an RP2350. The original drive's ROM served as a reference for protocol, status codes and timing.

**How it was built.** Every step from the first power-up of a bare mechanism to the board in your hands is written up, with scope traces, Apple screenshots and the mistakes, on the project page: bytehamr.com/projects/unidisk-controller. Section 10 of this manual is the short version, with a link to each log.

### TESTED ON

- Apple IIe with the Liron card (Apple 3.5 Disk Controller), ProDOS.
- 800K double-sided ProDOS disks: boot, read, write, format, catalog, copy.
- Daisy chain: Disk Hamr in front of a stock UniDisk 3.5, and behind one.
- Disk swaps, write-protected disks, empty drive, eject and reset in the middle of a transfer, long write runs (1610 blocks in one session, no errors).

### UNTESTED

- Apple IIc (with the UniDisk 3.5 ROM) and Apple IIgs (built-in disk port). Both talk the same SmartPort protocol, so it should work. It has never been plugged into either.
- Macintosh disks (524-byte blocks) and 400K single-sided disks. Out of scope by design.
- Any mechanism other than the Sony 800K double-sided unit the UniDisk 3.5 shipped with.

### WHO THIS IS FOR

Someone who has already decided the controller board in their drive is dead and wants the drive back. It assumes you can open the drive, handle a ribbon cable and read a connector's pin 1 mark. It does not explain what a floppy drive is.

SECTION TWO

# 2 In the box

- Disk Hamr rev 1.0, assembled and tested, firmware v1 loaded.
- One 20-way ribbon cable, IDC socket at both ends. It connects Disk Hamr's mechanism header (J2) to the mechanism's own 20-pin header. The IF-83's mechanism ribbon ends in a flat connector that does not fit a shrouded header, so this cable replaces it.
- This manual, as a QR code to the PDF.

Not included: a micro-USB cable. You only need one for a firmware update, and you already have one.




**FIG 2-1** *Box contents on the bench: board, ribbon, QR card.*

SECTION THREE

# 3 Before you start

## The drive's cables carry three power rails.

The DB-19 and the ribbon from it carry +5 V, +12 V and −12 V next to signal lines. The mechanism ribbon carries +5 V and +12 V. A ribbon plugged in backwards or off by one row puts a supply on a signal pin.

- Use the drive's own cables and the ribbon in the box. Do not improvise a DB-19 adapter.
- Match pin 1 on every connector before you push it home. Ribbon cables have a red or marked edge on pin 1; shrouded headers have a key notch; the mechanism header has a pin 1 mark.
- Power off the Apple before plugging or unplugging anything on the disk port.

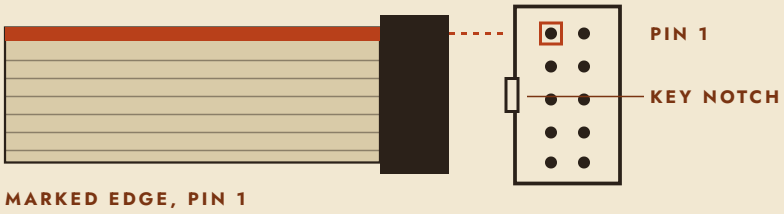



**FIG 3-1** *Ribbon cables have a red or marked edge on pin 1; shrouded headers have a key notch.*

During development a homemade DB-19 adapter put −12 V on a signal line (the log). It destroyed the IWM chip on the Liron card, which then had to be replaced with a donor. That chip is socketed and a donor was found, but it is the most expensive mistake available in this job and it is entirely avoidable.

**Static.** The board has a bare RP2350 module and CMOS level shifters. Touch the chassis before you touch the board.

SECTION FOUR

# 4 Is the board really dead?

Most "dead" UniDisks have dirty heads or a dry carriage rail, so start there: isopropyl on the heads, a drop of light oil on the rail, and try the disk again.

Here is how I knew mine were dead. With the drive plugged into the Liron card, the Apple wouldn't boot ProDOS from any disk at all, including a Disk II in another slot. The board was holding a line on the SmartPort bus and nothing could get past it. Unplug the drive and the Apple boots again. Two of my drives did exactly that, and cleaning changed nothing.

If you see the same thing, the board is the problem. It doesn't matter which chip on it failed, since the gate array in the middle of it has no datasheet and no replacement, and that's why this project replaced the whole board instead of repairing it (Going around the CXD1013).

If the Apple boots fine with the drive attached and the drive just never spins or never reads, the mechanism is the more likely suspect, and a new controller board won't change that.

SECTION FIVE

# 5 Replacing the board

Tools: a Phillips screwdriver. Nothing else.

1 **Open the drive.** Four screws on the underside, lid lifts off toward the rear.

PHOTO 5a

**FIG 5-1** *The drive on the bench, lid on, rear toward the camera, the four underside screws visible or called out.*

2 **Unplug four cables from the IF-83.** Front-panel LED lead (2 pins), eject button lead (3 pins), the ribbon from the front DB-19 (20 way), and the mechanism ribbon (20 way, flat connector). Note which way each one faces.

PHOTO 5b
This is the photo the whole section hangs on.

**FIG 5-2** *Lid off, the IF-83 in place with all four cables attached and labeled.*

## 5 Replacing the board, continued

3 **Unscrew the IF-83.** Four M3 screws into the standoffs. Keep the screws; Disk Hamr uses the same holes.




**FIG 5-3** *Old board out, standoffs visible.*

4 **Set the old mechanism ribbon aside.** It stays with the IF-83. The mechanism end of the new ribbon goes where it was: pin 1 to pin 1.




**FIG 5-4** *New ribbon seated on the mechanism header, pin 1 edge and the header's pin 1 mark both visible.*

**5** **Place Disk Hamr** on the standoffs, rear DB-19 through the case opening. Four screws. The outline and the rear connector were fitted with 3D prints before the board was made (outline, DB-19, populated board).




**FIG 5-5** *Disk Hamr screwed down, no cables yet.*

**6** **Plug in the four cables.** The silk on Disk Hamr labels each header by its reference; the two 20-way headers sit at the top edge of the board, the two small ones on the tab beside the rear DB-19:

- **J2**, top left: the new ribbon from the mechanism. Shrouded, keyed.
- **J3**, top right, under the rear DB-19: the ribbon from the front DB-19. Shrouded, keyed.
- **J5**, the 2-pin header on the tab: the front-panel LED lead. The shroud only lets it in one way.
- **J4**, the 3-pin header beside it: the eject lead. Same.




**FIG 5-6** *All four cables in, labeled J2, J3, J4, J5.*

**7** **Lid on, screws in.** Plug the drive back into the Apple.




**FIG 5-7** *The drive closed, cable on, back in its place beside the Apple IIe.*

## Time: ten minutes.

# 6 SECTION SIX First boot

Plug the drive into the Liron card's DB-19, or into the rear port of another drive. Turn the Apple on.

- The front LED lights while the drive is being accessed and goes out between accesses. It flickers through a boot and a catalog.
- `PR#5` (or whichever slot the Liron card is in) boots from the disk in the drive. ProDOS lists it as an 800K SmartPort device in that slot.
- The eject button ejects the disk, at any time, as the original did.

If the Apple does not boot and the drive is the only thing on the bus, go to section 8.

# 7 SECTION SEVEN Daisy chain

The rear DB-19 is a full SmartPort pass-through. Disk Hamr can sit in front of a stock UniDisk 3.5 or behind one; both orders are tested (the chain log). The Apple numbers the drives in cable order: the one plugged into the card is D1, the next is D2.

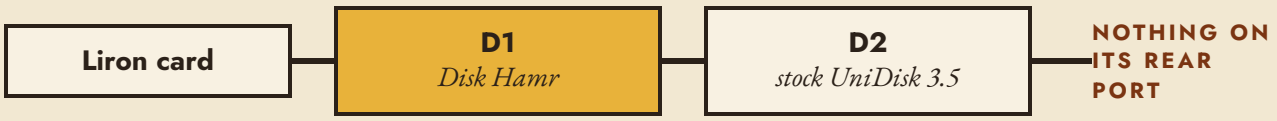



**FIG 7-1** *The one plugged into the card is D1, the next is D2.*

Chain rules are the same as for the original:

- The last drive in the chain has nothing on its rear port. Disk Hamr senses this (the LASTONE line on rear pin 16) and tells the Apple it is the last device.
- A drive further down the chain only becomes visible after the one in front of it has been initialized, which the Apple does at boot. Plugging a drive onto the rear port after boot needs a reboot.

# 8 SECTION EIGHT
# If it does not work

In order, stopping when it works.

1. **Apple will not boot at all with the drive attached.** Power off. Reseat J3 (the ribbon from the front DB-19) and check its pin 1 edge. Reseat the DB-19 cable at the Liron card.

2. **Apple boots, drive never spins.** Power off. Reseat J2 at both ends, pin 1 to pin 1. A reversed mechanism ribbon is the most likely cause of a dead-quiet drive.

3. **Drive spins, Apple says I/O error.** Try a known-good disk. Clean the heads. Try the disk in a working drive if you have one.

4. **LED never lights but everything works.** The LED lead is in J5 the right way around by the shroud, so check the LED board itself and its lead.

5. **Eject button does nothing.** Reseat J4. The button is a changeover switch; Disk Hamr reads the contact that closes when pressed.

Still stuck: open an issue on the repository with what the Apple shows, which slot the card is in and what else is on the chain.

`[REPOSITORY URL]`

SECTION NINE

# 9 Firmware

The board ships with firmware v1, the code that passed the bench milestones. It needs nothing from you, and the plan is that it never will.

If a later version does ship, know this first: updating means taking the board out of the drive. The Pico's USB jack and its BOOTSEL button face the inside of the case and can't be reached with the board on its standoffs, so an update is section 5 in reverse, the whole board out, then the swap again. There is no update port on the case and I'm not adding one; a drive from 1985 with a USB socket in the back is a different project.

With the board out:

1. Unplug the drive from the Apple before anything else.
2. Hold the BOOTSEL button on the Pico module, plug a micro-USB cable from the Pico to a computer, release the button. A drive called `RP2350` appears.
3. Drag the new `.uf2` file onto it. The drive disappears when it is done.
4. Unplug USB, put the board back in (section 5, steps 5 to 7), plug the drive back into the Apple.

If you work out a tidy way to do this without the teardown, a cable routed to a spare opening, a part, a bracket, send it. I'll link it from the project page.

Source, schematic and firmware: `[REPOSITORY URL]`. MIT license. Build it yourself with the Pico SDK; the README covers it.

# 10 SECTION TEN How it was built

Disk Hamr started as a Pico 2 and five level-shifter boards on a breadboard, wired to a mechanism from a dead UniDisk. It was brought up in nine milestones on the bench, each one logged with scope captures and Apple screenshots:

| | |
|---|---|
| **M0 and M1** | Power-up behavior of the mechanism, then status reads. |
| **M2 and M3** | Motor, seek and eject, then raw flux capture decoded on a Mac. |
| **M4** | Whole-disk reads on the board itself. |
| **M5** | Writes and formats. A disk written by Disk Hamr was read back by a working UniDisk, and the other way around. |
| **M6** | SmartPort on the RP2350. The Liron card's first packet, then ProDOS booting from a RAM disk through the card. |
| **M7** | ProDOS booted from a real floppy. Every block the Apple read through Disk Hamr was compared byte for byte against the same disk in a working UniDisk, with the disk in, write-protected and with no disk. |
| **M8 and M9** | Daisy chain with the working UniDisk, both orders, a full 800K disk copied across and booted from the copy. Then disk swaps, swap-then-write, write-protected disks, empty drive, eject and resets mid-transfer, 1610 block writes without an error. |
| **The board** | Routed, laid out again by hand, printed, fitted and ordered. |

The board itself was measured against a real IF-83: connector positions, the LED circuit (180 Ω pull-up, transistor sink), the eject switch wiring, and the rear-port pinout, all with a meter on the original board. The layout was printed five times and fitted in the case with the mechanism before any copper was ordered.

All of it, with photos, scope traces and the things that went wrong: bytehamr.com/projects/unidisk-controller.

# 11 SECTION ELEVEN Specifications

| | |
|---|---|
| **CONTROLLER** | Raspberry Pi RP2350A (Pico 2 module) |
| **LEVEL SHIFTING** | 5 × SN74LVC8T245, 3.3 V to 5 V |
| **HOST INTERFACE** | SmartPort over the Apple DB-19, Liron card |
| **MECHANISM INTERFACE** | Sony 800K 20-pin |
| **REAR PORT** | DB-19 daisy chain: PH3, /ENBL pass-through, LASTONE |
| **POWER** | +5 V and +12 V from the Apple through the host cable; no external supply |
| **BOARD** | 100 × 110 mm, 2 layer, fits the IF-83 standoffs and case opening |
| **FIRMWARE** | C on the Pico SDK, MIT license |

Schematic, board files, bill of materials and firmware source are in the repository.

# Disk Hamr

bytehamr.com/projects/unidisk-controller

REV 1.0, FIRMWARE V1

